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26

Ice Hockey



Matt Nichol

Performance Demands of Ice Hockey

While every sport has its own intricacies, I would argue that no sport requires such a robust combination of highly specific skills, variable energy system demands, and physical resiliency as the sport of ice hockey. The playing surface alone creates truly unique demands on the body unlike any other team sport.

In this section, the sport of ice hockey is described, including the various physical demands that it imposes on its participants. Additionally, we'll examine the role of the unique playing surface as well as the equipment and rules of play, which impact the way we need to prepare ice hockey players to be successful.

Sport Description and Factors of Winning

There are currently two distinct versions of the game of ice hockey: the International or Olympic version and the North American or National Hockey League (NHL) version. For the purposes of this chapter, we will be mostly referring to the North American or NHL version of ice hockey. Although there are very obvious differences with respect to the size of the playing surface and some slight variations in the rules, the energy system demands are similar. The most

significant difference between the two formats is the larger playing surface in Olympic-style ice hockey. This allows for a greater number of tactical options for coaches and requires its players to be more proficient skaters, or perhaps more accurately, exposes those players whose skating insufficiencies might be covered up through above-average size and strength or stickhandling abilities. The Olympic version of the game may require a slightly larger anaerobic fitness component, but the reader can extrapolate from the suggestions made herein and apply these to the international game as needed.

Ice hockey is played on a hockey rink, which is an oval sheet of ice contained within a perimeter of high-density composite plastic walls referred to as the boards. The playing surface is 61 m $\times$ 26 m (67 $\times$ 28 yd), with a corner radius of 8.5 m (9.3 yd), and the distance from the end boards to the nearest goal line is 3.4 m (3.7 yd). The boards surrounding the playing surface prevent the players from going out of bounds at any time; however, play is immediately stopped if the puck exits the playing surface (entering either the player's bench or the spectator seating area). The playing surface is made entirely of ice, which imposes unique physical demands on the players from both a biomechanical and physiological standpoint (see following).

A game consists of three periods of 20 min each, played with stop time, with the clock running only

when the puck is in play. An 18 min intermission occurs after the first and second periods, and the total average time of an NHL hockey game is 2 h and 30 min. Games are played between two teams of six players per side, including five skaters and one goaltender. The goaltender is able to play the puck offensively from directly in front of his net or inside the trapezoid area directly behind his net. Typically, the goaltender does not leave the immediate area of the goal or net. The five skaters on the ice are divided into three forwards and two defensemen. The forward positions consist of a center and two wingers: a left wing and a right wing. The defensemen play together as a pair and are divided into left and right depending on their dominant shooting sides. Forwards play together as units or lines, with the same three forwards consistently playing together. Teams will generally play each game with four lines of forwards and three pairings of defensemen and two goaltenders (starter and backup), for a total of 20 players dressed each game.

A substitution of an entire unit at once is called a line change. To achieve this, teams typically employ alternate sets of forward lines and defensive pairings when shorthanded or on a power play (down a man due typically to a penalty). Substitutions are permit-

ted at any time during the game. When players are substituted during play, it is called changing on the fly. When a team substitutes a player on the fly, the defensemen are forced to remain on the ice longer to protect against any scoring opportunities by the opposition while the team is momentarily outnumbered. As a result of this tactic, the defensemen end up having fewer opportunities to change and spend a greater amount of time on the ice. This point is important with respect to the time-motion analysis of the game and the subsequent additional conditioning demands for defensemen.

Relative Contribution of Physical Performance

Although ice hockey is a physically demanding sport, the primary differentiating factor between elite performers is not one specific measure of strength or fitness but rather high levels of technical skill with the puck, elite-level skating ability, and tactical awareness (commonly referred to as *hockey sense*). While it would be expected that all players at the professional level would exhibit levels of strength and fitness that would be superior to the amateur population, it is not uncommon to find elite performers in the sport who



Figure 26.1 The position of the ice hockey player on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation in ice hockey, acknowledging that this varies slightly between positions. Outside of critical psychological and team chemistry components, which are difficult to quantify, the pie chart shows the general relative importance of skills (40%), tactical awareness (30%), and physical capacities (30%) for ice hockey success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

have average or even below-average levels of strength or fitness compared to their cohorts. Figure 26.1 illustrates the relative importance of these factors in relation to other common sports found in this book.

Common markers of physical fitness do play a role in team success, but like many team sports, there is no direct correlation (11). While it stands to reason that faster and better-conditioned teams pose a greater chance of success, due to a lack of standardized testing across teams and the confidential nature of such tests for most teams, it is hard to quantify this with any degree of certainty. One longitudinal study published on an NHL team, however, showed that markers related to team success included peak 5 s anaerobic power performed during a 30 s Wingate power test, team average percent body fat, and lean body mass (12).

While the purpose of this section of the chapter is to discuss the physical preparation of ice hockey players and specifically to examine methods for developing the relevant energy systems, it must be stated that the number one determinant for success is hockey-specific skill. This is an important consideration for those responsible for the physical preparation of elite ice hockey players, as we must never lose sight of the relative importance of specific on-ice training for the athletes.

The literature available with respect to the anthropometric and physiological profiles of professional hockey players reveals that while these athletes are superior to the average recreational player, their profiles are not as strikingly dissimilar as might be seen in a professional basketball player, American football player, or elite track-and-field athlete. What sets the elite in ice hockey apart is their ability to technically apply their physical attributes on a surface of sheer ice balancing on a blade less than 1/8th of an inch thick. The act of skating on ice is a very unnatural movement and expert performance requires thousands of hours of technical training. Likewise, the ability to control the puck and make offensive plays with it (passing, scoring) requires incredible dexterity, exceptional visual processing skills, hand-eye coordination, and functional strength that is difficult to develop outside of playing the sport itself.

Ice hockey is generally regarded as the fastest team sport in the world. The players themselves can achieve speeds on their skates in excess of 40 km/h (25 mph). When shot, the puck can travel up to 180 km/h (112 mph) and punish players brave enough to place their bodies between the shooter and their goal. For goaltenders to successfully save

such a shot requires lightning-quick reactive abilities. The lack of friction on the ice surface coupled with the high speed of movement necessitates great eccentric strength to execute the repeated rapid stops and starts and changes of direction of travel that occur during each shift. Such movements additionally require high metabolic rates.

Low body fat percentage, high lean body mass, high relative strength (squat, chin- or pull-up) and power (Wingate peak and mean power, vertical and broad jump) have all shown high correlation with on-ice accelerative ability, but not necessarily individual point production or team winning (11). It is also important to note that ice hockey is a high-speed collision sport that incurs an extremely high incidence of injury during play. Those responsible for the physical preparation of elite-level ice hockey players must take this aspect into consideration when designing the annual training plan, by including posture-specific balancing exercises, skating strength, power, and agility exercises (9).

Targets of Physical Performance in Ice Hockey

As I've touched on, ice hockey is a high-speed, discontinuous, repeated-sprint sport. The play on the ice is characterized by rapid and irregular changes in movement patterns, accelerations and decelerations, as well as alternations between extremely high- and relatively low-intensity movements over a short time period (5). It is a sport that involves high-speed collisions and high-speed projectiles (pucks, sticks, and skate blades), and thus involves a large degree of potential for injury and bodily harm.

While some requisite level of aerobic fitness is likely to be beneficial for ice hockey performance, I don't believe you need to achieve this through traditional aerobic training that you might see in some of the other sports presented in this book. In my experience, athletes are prepared using almost no traditional aerobic training (outside of occasional walks in nature), yet these players can achieve some of the highest scores in their team's fitness testing rounds, play some of the most minutes on their teams, and in excess of 1000 NHL games over a career. While such outcomes are believed to be unachievable without a good level of aerobic fitness, I believe these results can be achieved using more unconventional but highly effective ways (see HIIT Weapons).

Forwards and Defensemen

Shift players may play for as long as 90 s or as short as 10 s, depending on a multitude of factors (penalties, poor line changes, coach decisions, official time interruptions, etc.). However, the average length of a shift for a forward in the NHL is 46 s, with a total time on the ice averaged at 16 min. The average shift length for a defenseman in the NHL is 55 s with an average total time on the ice of 20 min. The recovery time between shifts is 3 to 5 min for forwards and 2 to 4 min for defensemen. Clearly, the nature of the game of modern ice hockey itself resembles a high-intensity interval training (HIIT) format, specifically the repeated-sprint training (RST) format. A time-motion analysis study by Jackson (7) suggests that the average intra-shift work:rest (W:R) ratio (purposeful medium- to high-intensity movement versus nonpurposeful or stationary to low-intensity movement) across all players during even-strength play was 1:1.6, and the game W:R (time on ice versus time on bench) was 1:3.7 (7). Each on-ice shift is composed of a series of rapid accelerations and decelerations of only 1 to 3 s, interspersed with low- to medium-intensity gliding and occasional physical struggles with an opponent for possession of the puck. Thus, the ability to perform a multitude of short explosive bouts of work with little recovery, known as repeated-sprint ability (6), and described within the book as RSS, or repeated-sprint sequences, is a critical requirement for ice hockey. This form of training (i.e., RST, chapters 4 and 5), and specifically type 4 and 5 targets,

needs to be the predominant training applied in an ice hockey player's program.

Time-motion analysis by Bracko et al. (3), illustrated in figure 26.2, shows that, contrary to popular belief, the majority of the time a player is on the ice, he is engaged in very low-intensity movements, such as gliding, low- to medium-intensity forward skating, and stationary play, both with and without the puck (3, 7). Thus, regardless of the position played, all skaters spend the majority of their time on the ice (up to 55%) performing low-intensity work. This time is important to allow the replenishment of energy stores required for subsequent high-intensity moments of play (see also chapter 4 (10)). As described by Montgomery (10), a typical player performs for 15 to 20 min of a 60 min game, where each shift lasts from 30 to 80 s, and there is typically 4 to 5 min of recovery between shifts.

Due to the lack of friction on ice, players continue moving after they stop active locomotion, and are able to glide with little to no work once their momentum is generated, making this sport low demand relative to others from an endurance context (figure 26.1). Conversely, stopping completely or making a 180-degree change of direction creates large mechanical costs relative to similar movements executed in land-based sport surfaces of high friction (court, field-turf, etc.). This aspect creates large neuromuscular demands on both forwards and defensemen, however, these demands may be greater on the defensemen as their movements also require a large reactive component.

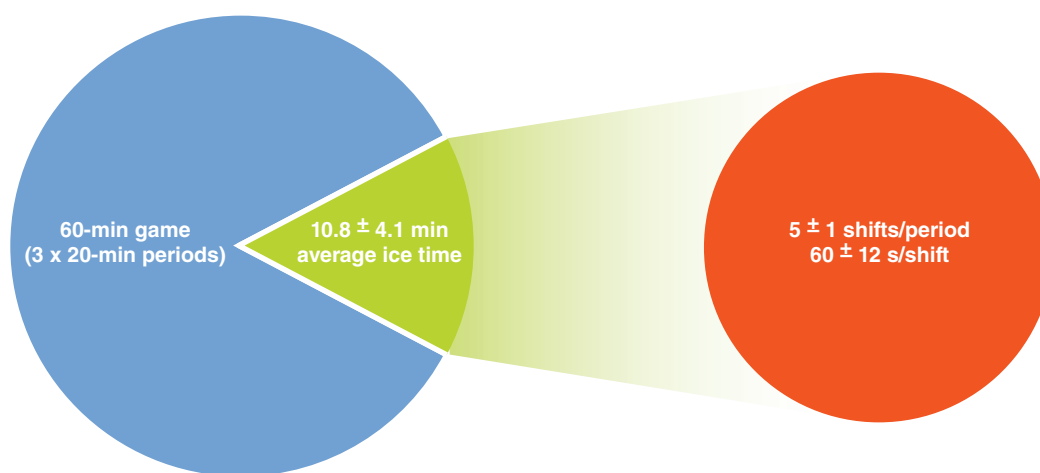


Figure 26.2 Time-motion analysis data showing both the mean ice hockey shift interval alongside the typical skating characteristics of the first 30 s of each shift period.

Adapted from M.R. Bracko, G.W. Fellingham, L.T. Hall, et al., "Performance Skating Characteristics of Professional Ice Hockey Forwards," *Sports Medicine, Training and Rehabilitation* 8(3): 251-263.

Nevertheless, the aerobic oxidative system is certainly active during ice hockey. With respect to its metabolic requirements, while direct $\dot{V}O_2$ data is limited, Bell and colleagues (1) reported that male forwards showed mean and peak heart rate (HR) values of 96% and 100% HRmax during competition, respectively, which recovered to 75% HRmax between shifts. Defensemen were similar, with work values of 92% and 96% HRmax, respectively, and recoveries to 71% HRmax between shifts. This data highlights the maximal intensity nature of the sport for forwards and defensemen.

The anaerobic glycolytic system is also engaged during ice hockey, although perhaps not to the degree that most in the sport believe it to be. For example, Nichol and McNeely (unpublished observations) measured lactates during a regular season game across periods in 10 male professional American Hockey League players (25 ± 2 yr, 90 ± 5 kg) (198 ± 11 lb) with NHL experience. Blood lactate increased from 2.4 ± 0.5 to 6.4 ± 3.7 mmol/L in period 1, from 3.4 ± 3.1 to 7.5 ± 2.8 mmol/L in period 2, and from 3.4 ± 1.9 to 7.1 ± 3.1 mmol/L in period 3. While lactate likely increased to a greater value than that measured during play (due to the lag time between exercise and measurement), it nevertheless recovered to homeostatic levels from one period to the next without any evidence of carryover effect.

Goaltenders

There is relatively little research published on the physiological demands of goaltenders. Each team generally has two goaltenders dressed to perform, and barring an unforeseen injury or substandard play, only one of these goaltenders will typically play during the match. Other than during complete stoppages of play (team or official time-outs, TV time-outs, or intermissions), the goaltenders do not have the luxury of a passive recovery period on the bench like their teammates do. Goaltenders must then have the ability to remain calm and focused for long periods of time, before displaying fast reaction speeds and explosive power at a moment's notice.

Goaltenders are truly aerobic/alactic, or RST-type athletes, with the majority of their time on ice spent standing in their crease waiting for play to develop. When required, however, a split-second reaction followed by an explosive and dynamic movement is needed to stop a shot. Goaltenders additionally require high levels of flexibility and, in particular, hip mobility, as well as high levels of core stability

and specific endurance of postural muscles to endure the ready position, with a flexed torso, for a substantial time. Goaltenders also must be able to play from a kneeling position.

Key Weapons, Manipulations, and Surveillance Tools

In this section, I present the different HIIT weapons used in ice hockey, their manipulations, along with ways we can monitor their effectiveness (surveillance). The breakdown of the different HIIT weapons used in elite ice hockey is shown in figure 26.3.

HIIT Weapons

The principle of specificity within sport is well documented. The more similar the movement patterns and work:rest ratios of the training exercises to the actual sport, the greater the transfer effect may be. Anyone who has experience playing hockey or training those who do will tell you that there is simply no substitute for on-ice work. Therefore, my belief is that the following components should form the majority of weapons involved with training ice hockey players:

- High-intensity alactic-dominant efforts (short-duration maximal efforts)

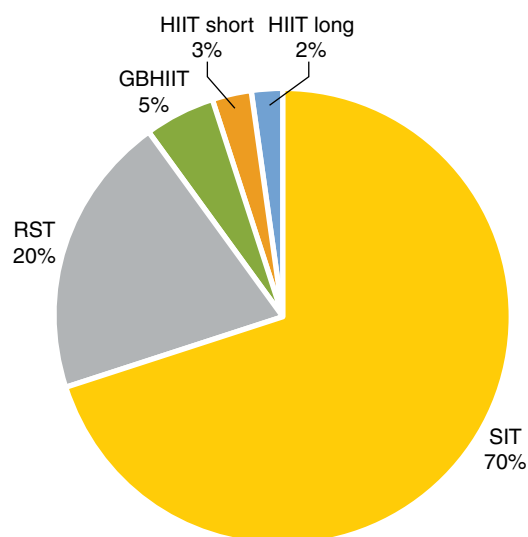


Figure 26.3 Percentage of the different HIIT weapons (formats) used throughout the annual season in elite ice hockey. HIIT: high-intensity interval training; RST: repeated-sprint training; SIT: sprint interval training; GBHIIT: game-based HIIT.

- Closed kinetic chain and weight-bearing exercises
- Repeated rapid accelerations and decelerations
- Unplanned and reactive movements
- On-ice specific exercise where possible
- Movements both with and without the puck
- Work:rest ratios of 1:2 to 1:3 for defensemen and 1:3 to 1:5 for forwards

Using these principles, the predominant weapons that form in ice hockey (figure 26.3) would be described as RST, SIT, and short-interval HIIT, mostly all done in a sport-specific format. Off the ice, long-interval HIIT is done occasionally, for specific situations described later, but rarely if at all for most players. As my study and practice in this sport has progressed over the years, I have realized long-interval HIIT is much less specific and beneficial than many believe it to be. While such training might provide benefit with regard to mental strength and cardiovascular endurance, by pushing a player to build a certain level of grit by accomplishing difficult and nonenjoyable tasks, I believe such training has a low carryover effect to game performance. The majority of our long-interval HIIT work is devoted to preparing our players for the training camp fitness testing that they will have to perform with their teams rather than for performance in the actual sport itself. Regardless of whether or not I agree with the specificity of the tests that these teams select, I do have an obligation to prepare my players so that they can perform well in these tests. Some of my more veteran and well-established players do not concern themselves with this type of training, for they know that they will not be judged on their performances at training camp. For younger players, however, or those who are less established, there is still a need to perform well on the team testing so as to make a statement to the coaches and management that they are hard-working, fit players.

Manipulations of Interval Training Variables

In the following section, I describe in detail the primary HIIT weapons used to train ice hockey players, as well as their manipulations. As mentioned, these include sport-specific RST, SIT, and short-interval HIIT. Long-interval HIIT is used sparingly in the preseason for certain isolated situations, and as these are so infrequently used for training ice hockey

players, they will not be elaborated on. Note that nearly all HIIT weapons are on-ice operations, and generally involve handling the puck, shooting the puck on goal, and tracking the opposing player. However, as there is no real play element involving decision-making, these are not classified as GBHIIT or SSG.

Sprint Interval Training

As shown in figure 26.3, SIT is the predominant form of HIIT used in elite ice hockey preparation in my experience. SIT for ice hockey can take many forms, but the predominant variations of SIT that I use involve five different drill formats that include cross-ice tracking, long-pass tracking, and chase-the-rabbit tracking drills.

Note that all ice-based drills are a collaboration between myself and Steve Spott of the San Jose Sharks. Steve is a good friend and is one of my on-ice coaches for the NHL off-season camp that I run.

Repeated-Sprint Training

In a discontinuous repeat sprint sport such as ice hockey, high-intensity maximal power might not be the only quality that you need to train, but it is certainly most important. Nobody cares how long you can skate if your skating is slow and your play ineffective. My philosophy for developing a player's capacity here is as follows:

1. Develop a high level of relative strength and power in the weight room.
2. Develop a high rate of acceleration power and deceleration strength on the track and on the ice.
3. Increase repeat sprint ability and repeated sprint ability on the ice.

In layman's terms, I believe we have to improve ability to accelerate explosively and decelerate rapidly, and do this over and over and over again. I believe that RST is instrumental in achieving this (i.e., HIIT type 4). Like many of the sport-specific SIT drills already discussed, the four-corner tracking drill is similar in terms of sport specificity, but in the RST format, the emphasis for this drill is really on explosive-quality skating.

HIIT With Short Intervals

On- and off-ice short-interval HIIT work is used more for warm-up purposes using the assault 10/20, the fox and rabbit off-ice tag game, and the on-ice fly

30-15 INTERMITTENT ICE TEST

Martin Buchheit

While yet to be adopted across NHL teams, one of the tests that has recently emerged is that of the 30-15 intermittent ice test (30-15_{IT}) (2, 4). Modeled after the running-based 30-15 intermittent fitness test (<https://30-15ift.com>; see chapter 2), the 30-15_{IT} has a similar duration of exercise and recovery period (i.e., 30/15 s for exercise/recovery) and shuttle distances (i.e., 40 m, figure 26.4), which can be used for assessing both cardiovascular fitness and anaerobic speed reserve specific to ice hockey. Starting speeds of 10.8 km/h (6.7 mph) for the first 30 s stage, together with a speed increment of 0.63 km/h (0.39 mph), is generally considered appropriate for reaching both a high HR and a fast skating velocity, while at the same time enabling exercise to last at least 5 to 6 min. The 30-15_{IT} can be set up in less than 5 min, and requires a measuring tape (2 × 20 m) (2 × 22 yd), 9 cones, a sound system with the 30-15_{IT} audio track, and at least two observers.

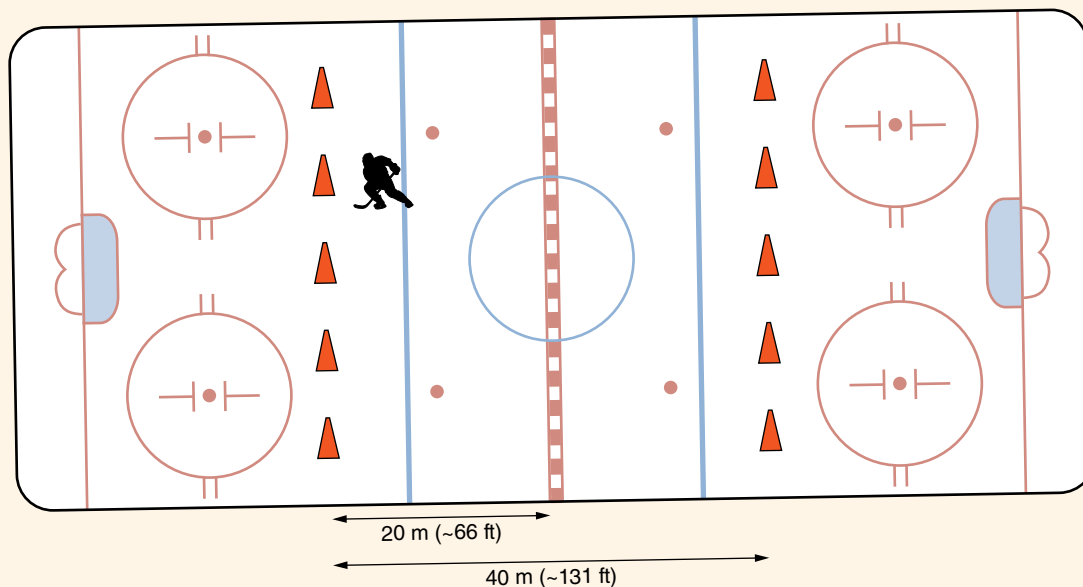


Figure 26.4 Area prepared for the 30-15 intermittent ice test (30-15_{IT}) (4).

As with the 30-15 IFT, an audio signal sounds at appropriate intervals for players to adjust their skating speed. At the start of each stage, players rest upright with their front skate parallel to the starting line of the 40 m shuttle course. At the first audible signal, the player begins skating while holding the hockey stick with the preferred hand. Skating velocity is adapted to meet the midpoint of the course (20 m) in synchrony with the second audible signal. Without stopping, and if necessary adjusting skating velocity, the player continues skating to reach the end of the course in synchrony with the third auditory signal. At the 40 m (44 yd) end, the player immediately changes direction and starts skating in the opposite direction (stop and go) to reach the midpoint in time with the fourth audible signal, and so on until the end of the 30 s stage.

As with the running version of the test, attention is paid to the change of direction on the extremity lines, with players needing to decelerate, cross the line with at least one foot, turn, and recommence skating. At the end of the 30 s stage, players are instructed to either glide or skate slowly to the closest line (either on the end of the 40 m [44 yd] shuttle course or on the midpoint line) for a 15 s rest period, before beginning the next stage. Similar criteria are used to determine the maximal skating speed (VS_{IT}). The test has been shown to be valid (i.e., final velocity on ice very largely correlated with the velocity reached during the off-ice test version, $r=0.72$) and reliable ($CV < 2\%$) (4), and may provide a more specific assessment of the anaerobic and neuromuscular demands of ice hockey, as evidenced by the higher blood lactate response in the 30-15_{IT} compared to the 30-15 IFT (2).

On-Ice 45 s Shift Length Conditioning Drill

The setup for the 45 s on-ice shift length conditioning drill involves dividing the team into four groups. Players start on the bench, before jumping onto the ice to skate easy/relaxed around for 30 s. On the coach's whistle, players perform 15 s of position-specific all-out sprinting (figure 26.5), followed by a hard line change

on a finishing whistle, whereby the next group exits the bench to commence the start of the drill. The goaltenders will work with their goalie coach during the 15 s phase to perform their position-specific movements. This is an all-out SIT drill with an effort-to-pause ratio of ~20 s on to ~3.5 min off, aiming for type 5 targets.

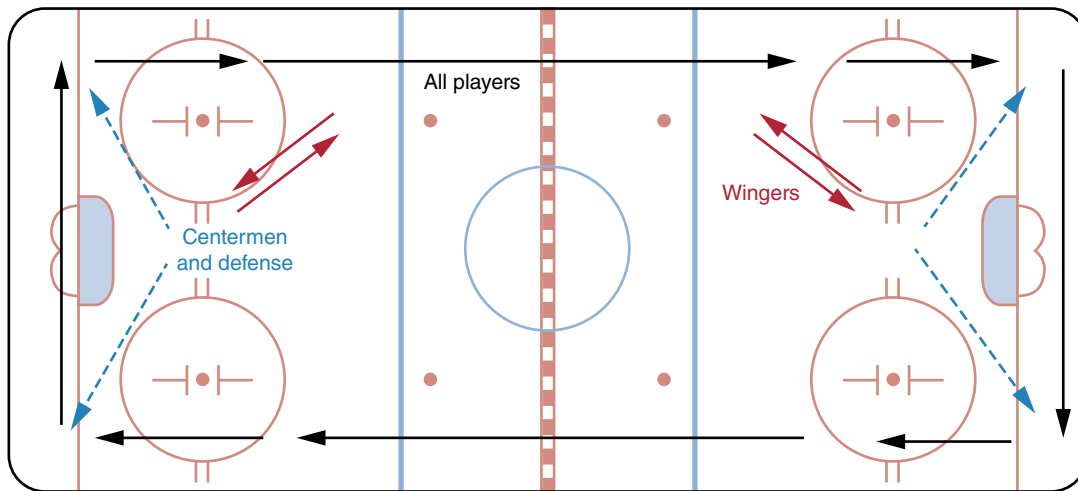


Figure 26.5 On-ice 45 s shift length conditioning.

drill. These are used sparingly in the off-season for specific players. The frequency can increase for some athletes whose fitness levels are subpar, for players who are not receiving regular shifts (those who are scratched from the lineup or playing a limited role on the fourth line), or players returning from a prolonged injury. In these latter cases, these short intervals can be used 3 to 5 times per month in-season to maintain conditioning levels.

HIIT With Long Intervals

Like short intervals, long intervals are used sparingly and really mostly in isolated cases when athletes have clear high-intensity aerobic conditioning deficiencies that require a solution or might have an injury and still require conditioning. The modality for HIIT with long intervals is generally nonspecific, including bike ergometer in the form of long-interval sets. One example of a long-interval session that I occasionally use in these isolated instances might be 3 min easy (RPE = 10), 2 min moderately hard

(RPE = 16), and 1 min maximal (RPE = 18 to 20), repeated 4 to 6 times.

Additional Strength and Conditioning Exercises for Ice Hockey Players

Most people (athletes and coaches combined) separate the work done in the weight room with barbells, dumbbells, cables, and so forth from any other work done. We tend to think of any work done with weighted implements as being solely strength or power training and assume that *metabolic conditioning* refers only to the training performed without weighted implements for specified periods of time or with specific heart rate goals. I do not believe this should be the case. As I've described, ice hockey consists of very short-duration bouts of explosive power and relies on the ability to rapidly accelerate and decelerate, regardless of the manner in which this occurs. Thus, I try to create similar demands on the athletes from the strength and power work we do in the gym.

Cross-Ice Tracking Drill

The cross-ice tracking drill is an ice hockey-specific SIT drill set up as shown in figure 26.6. Here, roughly five players begin in each of the four corners of the rink. Starting at one end and one side of the ice, on the coach's whistle, the player with the puck skates straight down the ice to shoot on net. An opposing player from the

other side of the ice crosses over to the opposite player on the whistle to track against him. For this drill, we ask for all-out speed and explosiveness and full energy, from both the puck carriers and trackers, with each player going through about four times, making an effort-to-pause ratio of about 20 s on, 2 min off, aiming for type 5 targets.

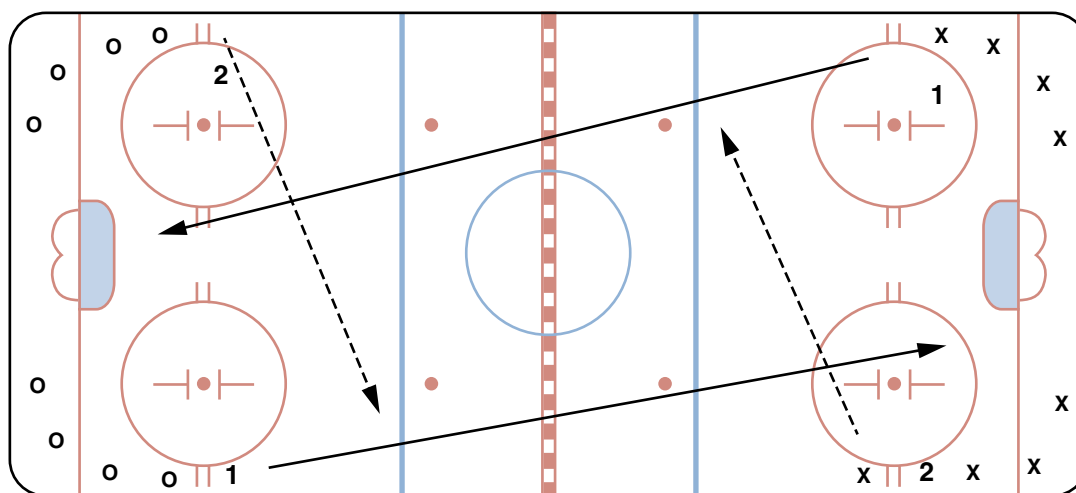


Figure 26.6 Cross-ice tracking drill.

When coaches think about the best methods of developing power in the gym, it is often the Olympic lifts and their variations that come to mind. Although Olympic-style weightlifting can be considered a sport unto itself, the exercises and their variations have been used by a wide variety of athletes outside of weightlifting to improve sport performance. Although the exercises do not mimic the specific actions seen on the ice, they do incorporate some of the more important aspects, such as being performed in a closed kinetic chain environment as total body lifts. With respect to the hang variations of the snatch and clean, the lifts begin and end in an athletic posture that is common to most team sports including ice hockey. Training these lifts can improve strength, rate of force development, and movement velocity. Another equally important benefit (or arguably more important) is the improvement of eccentric strength and rate of force development, which greatly enhance the ability to decelerate and change direction at high speeds during game play.

Although there are tremendous benefits to the use of Olympic-style weightlifting, there are also many inherent problems for ice hockey players. Many ice hockey players lack the requisite joint mobility to perform the lifts properly. The nature of the sport of ice hockey requires players to be in a flexed-torso position, which affects both hip and thoracic spine posture over time and leads to difficulty with overhead lifting. Many ice hockey players at the professional level also have difficulty with the catch position of the lift as it places tremendous strain on the wrists. Stickhandling in ice hockey is a highly specific skill and induces repetitive strain of the wrists in a flexion pattern. Injuries to the wrist are highly problematic at the elite level, since hockey players require sound wrists for stick handling, passing, and shooting of the puck, and such fine-motor skill can be disrupted by even the smallest interruptions in feel. As such, players are normally very tentative about performing any nonspecific activities that they believe might cause excessive

Long-Pass Tracking Drill

With the long-pass tracking drill (figure 26.7), players start at one end on opposite corners, with roughly five players in each of the four corners. On the coach's whistle, players make two long passes (one each) before (again on the coach's whistle) taking a shot on net, then turning hard to track back against another player. For this sport-specific SIT drill, we ask for all-out speed and energy, and each player goes four

times. The defending tracker of the puck carrier must track back through the middle of the ice with good stick position, and players must execute long passes, working on pass accuracy and receiving. All-out speed and explosiveness here is emphasized for both trackers and puck carriers, making an effort-to-pause ratio of about 20 s on, 2 min off, aiming for type 5 targets.

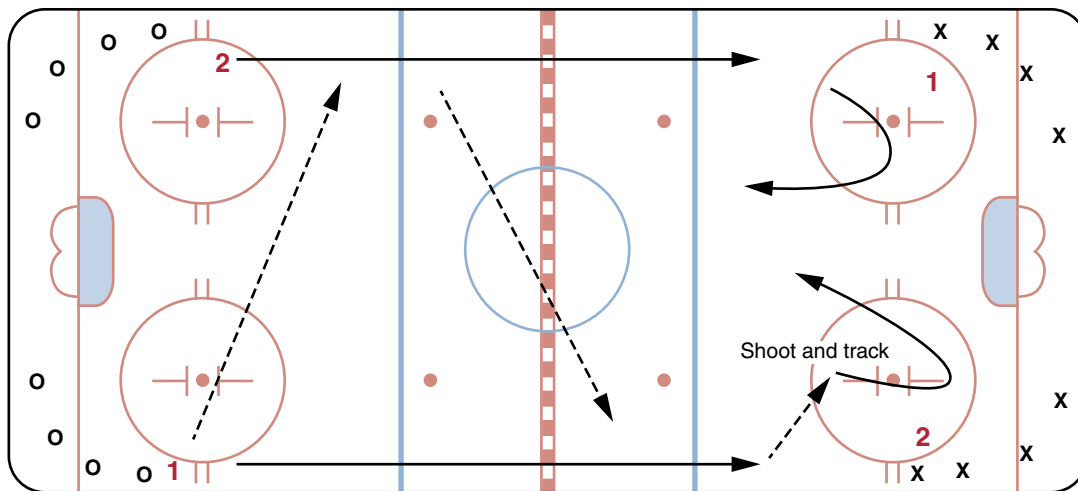


Figure 26.7 Long-pass tracking drill.

strain at this joint. For athletes who find the Olympic lifts and their variations to be problematic, alternate exercises include the second pull of the Olympic lifts without a catch, the use of loaded jumps, resisted sprints, and structuring our training program using conjugated periodization or a modified “Westside” style of conjugated periodization programming.

Additional conditioning exercises I use for ice hockey players are described in table 26.1. The skipping complex is done daily, but only as a warm-up with 2 repetitions. The slideboard exercise is done once a week in June when we are not skating, but more as prehabilitation or part of a rehabilitation program for any injured athletes. For this, we focus on controlled movement, not all-out conditioning. The sled work is done once per week in late June and early July, while the medicine ball toss and chase drill is done a couple times a summer as a fun finishing exercise or as a punishment to incentivize athletes to work hard and focus on a previous drill.

Recovery Programming and Adjunct Training

There are additional activities that I feel are important to support the high-intensity nature of the on- and off-ice training applied in ice hockey players. These include walking meditation, yoga, Qigong, and Pilates. All of these activities are quite different, but the common denominator is that they teach the athletes to connect proper breathing patterns with movement and, when done properly, will stimulate the parasympathetic nervous system, which is beneficial for recovery.

Load Surveillance and Monitoring Tools

When I worked with the NHL's Toronto Maple Leafs hockey team, we used heart rate monitoring during every practice and discovered that many of the drills described (figures 26.5 to 26.10) achieved much

Chase-the-Rabbit Tracking Drill

With this ice hockey-specific SIT drill, shown in figure 26.8, we start with roughly six forwards and four defensemen at opposite ends. On the coach's whistle, players skate all out (full energy) down the ice, 2 on 1, with a puck to pass and shoot on net before turning hard back to track against the opposing players performing the same action. Each player goes four times before switching sides and repeating. The game ele-

ment of chase the rabbit is mostly for the defending player who is tracking the puck carrier. The defending player should be holding a good gap between the 2 on 1 against him, knowing he has a tracker applying backpressure. We ask for all-out speed and explosiveness for both the puck carriers and trackers, and this drill tends to result in an effort-to-pause ratio of about 20 s on and 2 min off, aiming for type 5 targets.

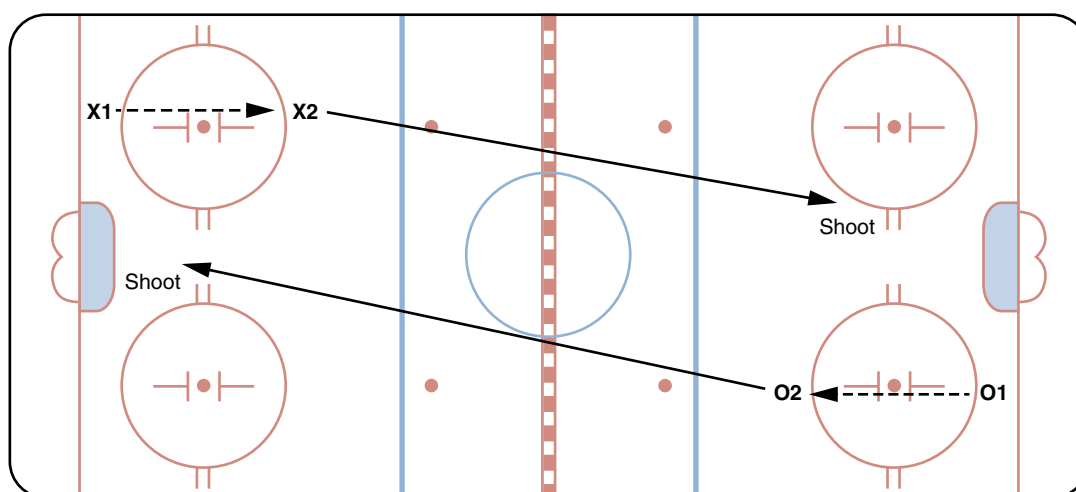


Figure 26.8 Chase-the-rabbit tracking drill.

Table 26.1 Additional Conditioning Exercises Used for Off-Ice Hockey Player Strength and Conditioning

Drill/exercise	Description	Repetitions
Slideboard intervals	15 to 30 s work, 30 to 60 s rest	6 to 10 reps
Sled push/pull	15 s reverse sled walk, 15 s forward sled push, 30 to 60 s rest	4 to 8 reps
Skipping complex	20 double leg jumps, 20 alternate leg double contact jumps (left, left, right, right), 20 hip twists, 20 high knees, 30 s rest	Repeat 5 to 8 times. If athlete can perform 8 sets of 20 reps each, increase to 5 sets of 30 reps each.
Medicine ball toss and chase exercise	Explosive medicine ball chest toss then sprint after the ball, two-foot stop at the ball and repeat. Upon reaching 100 m (109 yd), collect the ball and sprint back to the start. Rest 30 to 60 s.	Repeat 2 or 3 times.

higher heart rates than expected. My guess is this might be due to the higher neuromuscular demand imposed. I would also measure the player's heart rate variability (HRV; chapter 9) following each game and the morning after each game to assess their recovery

and readiness to train. My team also performed comprehensive blood and urine analyses before, during, and after the season to examine effects on the player's biochemistry. While the data collection was very rudimentary, it was still informative. Today

Four-Corner Tracking Drill

With the four-corner tracking drill, shown in figure 26.9, players on all four corners of the rink (shown as Xs and Os, 1 to 4) skate down the ice with a puck and on the coach's whistle take a shot on net before making a tight turn to track back against another player. Roughly five players begin in each of the four corners, and each player goes through the drill three times.

For this drill, we demand full all-out speed and energy from the players. While we ask for an accurate shot on goal by the offensive player and purposeful tracking by the opposing player, we are really looking for speed and explosiveness. The effort-to-pause ratio for this drill is roughly 12 to 15 s all-out, 45 s rest, aiming for type 4 targets.

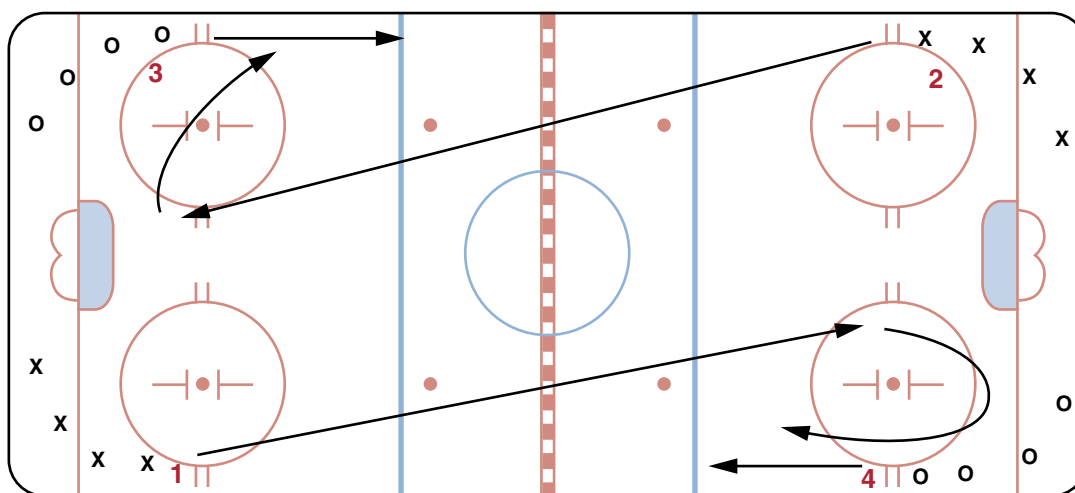


Figure 26.9 Four-corner tracking drill.

Assault 10/20

The assault bike is a stationary bike that uses wind resistance by means of a large fan wheel that matches the athlete's power output with an equal amount of resistance. This enables athletes to easily perform HIIT work without having to fumble around with buttons or dials, as the machine automatically scales the workload according to the player's capacity as needed on that particular day. Our go-to HIIT cycling sprint workout involves a 4 min bout of 10 s hard (RPE starting at 15 and building to 20) interspersed with 20 s of recovery (passive spinning). Objectively the heart rate data that we collect shows that this hits type 1 targets and can improve Wingate peak and mean power scores. Subjectively, players will routinely

tell us that this test mimics the physical feeling they get from a preseason conditioning skate practice.

We typically use the assault 10/20 once per week in the second phase of the off-season training period (late June to mid-July) to build a player's fitness. In the final phase of the off-season (late July to late August), we require the athletes to successfully complete a certain amount of work on this test on at least two occasions before we trust them to maintain their fitness in the off-season, and thereafter we no longer test them. For the assault 10/20, which would be considered a short-interval format, we aim for 5 to 8 min of total work time broken down into 1 min sets with 3 to 4 min recovery between series.

Fox and Rabbit Off-Ice Tag Game

We use a variety of methods to incorporate vision, reaction, coordination, and high-speed decision-making into our drills, which include but are not limited to visual and auditory cues by the coach and reactive light devices (fusion sport, Fit-Lights, etc.). None of these are as effective, as fun, or as efficient for team training as a good old-fashioned game of tag. Tag does not require any expensive equipment and can be administered to very large groups by a single coach with little more than a stopwatch and a whistle. We use a variety of different formats depending on the size of the group or the primary objective of the workout.

One example of a particularly effective version that we use for groups of 12 to 20 athletes is described. Note that when there are fewer athletes you can reduce the size of the playing area or use only one active chaser (fox). In a

$30 \times 30 \text{ m}^2$ ($33 \times 33 \text{ yd}^2$) area marked off by pylons, two players are designated as being foxes while the remaining athletes are rabbits. The goal of the foxes is to attempt to tag as many rabbits as possible in 30 to 45 s, thus resembling the short-interval HIIT format (types 3, 4). When players are tagged, they immediately stop where they are and perform an isometric push-up. Players who have not been tagged by the fox can free a rabbit who is holding an isometric push-up by jumping over the rabbit. The drill continues for 30 to 45 s and then the players have 30 to 45 s of active rest during which they must keep walking. Players get no early warning of when the next work bout will begin, so they must keep their wits about them during the rest period and be prepared to react. The two foxes with the greatest number of rabbits in an isometric push-up at the end of the active time are the winners.

Fly Drill

For the fly drill, illustrated in figure 26.10, players on opposite sides (shown as Xs and Os) skate hard down the ice with a puck and shoot on goal. On the coach's whistle, the player returns by tracking back the opposing player. For this drill, we have roughly 8 to 10 players in each opposite corner, and have each player go through this three times before switching sides and repeating the drill. The fly drill is done at

about $3/4$ maximal skating speed effort, and is done continuously without stopping once the drill has commenced. With the exception of taking a shot on an empty net, there's not really any game element to this drill. The effort-to-pause ratio is about 10 s on ($3/4$ maximal skating speed effort), 20 s off, and the fly drill is typically used as a warm-up drill, hitting type 2 targets.

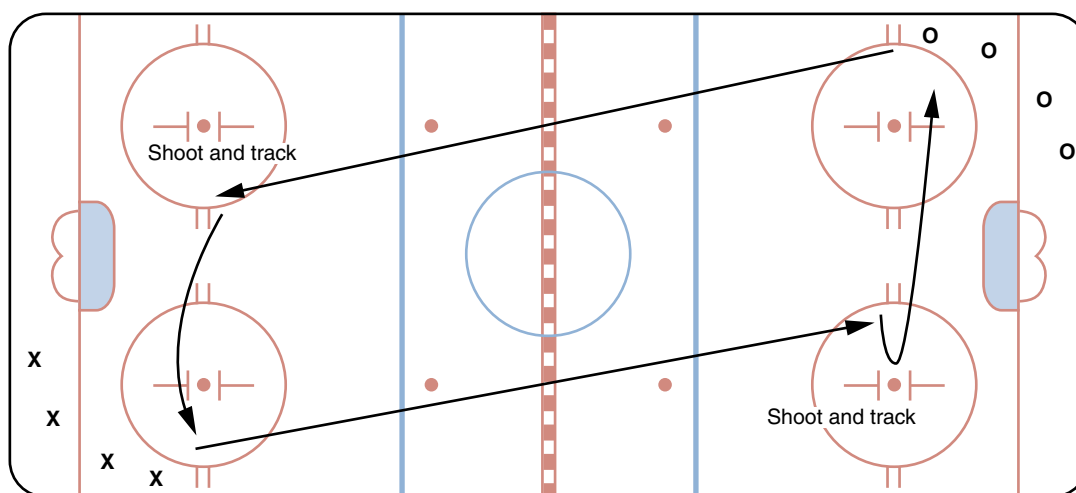


Figure 26.10 Fly drill.

Walking Meditations

There is a great abundance of research that shows that a regular meditation practice can improve focus and concentration, reduce stress, enhance sleep, and improve overall well-being. There is research to show that even as little as 10 minutes per day can be effective. Despite this, many athletes are reluctant to make this a structured part of their training regimen.

I encourage all of my athletes to incorporate meditation into their daily routines, and I have

found that one of the most effective ways to do this is with walking meditations.

I encourage all of my athletes to walk at least one day per week for a minimum of 40 min. They are asked to walk at a comfortable pace and focus on controlled-breathing patterns. I never track these sessions with regard to intensity or look for a specific effect, but I certainly feel that this is important to control stress and restore the player (see chapter 7).

Yoga and Qigong

In the second phase of our off-season (middle of the off-season), our yoga and Qigong work becomes more flow-based, with less rest between postures. Most of our athletes will

average around 50% to 60% HR_{max} during these sessions, but less-conditioned athletes or those with particularly poor posture and mobility can display even higher values.

my surveillance tends to focus more on direct observation of the athlete's behaviors, alongside direct communication with the athlete, although we do regularly use the following pieces of technology to monitor progress:

- Heart rate monitoring
- Accelerometers to measure bar speed velocity in the weight room
- Nord Board to assess both hamstring strength and pelvic asymmetries
- Laser timers to test acceleration and speed both on and off the ice
- Complex Electrical Muscle Stimulation to assist with recovery from training and injury rehabilitation

Strategies for Structuring the Training Program Using HIIT

The following section outlines strategies for structuring the HIIT aspect of the training program for elite ice hockey players.

Controllable and Uncontrollable Factors

As with most of the professional sports highlighted in the book, ice hockey conditioning is mostly limited to the work that can be done in the preseason to prepare for team testing. Once the season commences, we are at the mercy of the 82-game regular season schedule, which generally means playing 2 to 4 times per week throughout an 8 mo period, before a similar rate of game playing when an athlete's team makes the playoffs. For the regular season then, alongside the significant travel aspect (41 away games throughout North America), the emphasis is on recovery for the key players and conditioning maintenance for benched, injured, or less-used players. A general program structure is shown in tables 26.2 and 26.3.

Sample Training Programs

The final section in this chapter details the yearly structure and progressions of the HIIT work across the season in elite ice hockey players. An example of a weekly training microcycle used during the general preparation phase is shown in table 26.2. For an NHL player this is typically in late June or early July. Note in this example how the program-

Table 26.2 Example of a Weekly Training Microcycle During the General Preparation Phase

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Focus	Dynamic effort lower-body strength training and RST	Max effort upper-body strength training and short intervals	Short intervals and recovery	Max effort lower-body strength training and RST	Torso and strength endurance	General aerobic	Rest
HIIT	RST contrast sprints: 5-7 × 14-23 m (5-8 × 15-25 yd loaded/20-30 yd unloaded, 30 s rest between loaded and unloaded phases (type 5) RST change of direction/agility drills: 3-6 s on, 15-30 s off × 10 (type 5)	Short intervals assault bike 10/20 test (type 1)	Tempo run warm-up 10 min easy. MS: 10 × 90 yd in ~15 s, 30-45 s rest (type 1)	RST 20 yd prowler push forward/reverse, 10 s on/60 s off × 5 each (type 5)	Short intervals assault bike 10/20 test (type 1)		
Strength	Dynamic box squat 8-12 sets of 2-3 repetitions, 55%-70% 1RM at >0.8 m/s, 3 min rest between sets	A1) Single-arm DB snatch, 4 × 3 at >1.0 m/s B1) Incline DB press, 4 × 4-6 B2) Dynamic cable wood chop 4 × 6-8 B3) Single-arm DB row 4 × 6-8 C1) Isometric neck strengthening, 3 × 6-10 × 6 s per position C2) Rolling side bridge and prone bridge and reach, 2 × 2	Yoga	A1) Trap bar DL, 4 × 4 A2) Band resisted standing long jump, 4 × 3 A3) Nordic hamstring curl, 4 × 3-5 B1) Single-leg skater squats, 3 × 6-10 reps B2) Single-leg RDL, 3 × 6-10 reps B3) Slideboard lateral lunge, 3 × 6-10 reps C1) Side bridge, 2 × 6 × 6 s	A1) Half kneeling cable chop, 3 × 7-10 A2) Half kneeling cable lift, 3 × 7-10 A3) Parallel grip chin-up, 3 × 7-10 B1) Swiss ball mountain climber, 3 × 40-60 s B2) Swiss ball stir the pot, 3 × 40-60 s B3) Swiss ball oblique lateral flexion, 3 × 8-12 C1) Wrist roller, 3 × 40-60 s C2) Isometric neck strengthening, 3 × 6-10 × 6 s per position		

Note the programming of the different HIIT types across the week to maximize adaptation, minimize interference, and accommodate for the neuromuscular demands. The consecutive type 1 HIIT on Tuesday and Wednesday would be likely to limit neuromuscular overload (Tuesday) and facilitate recovery (Wednesday), allowing the hardest work (type 5) to be performed on Thursday.

ming of the different HIIT types across the week are placed so as to maximize adaptation, minimize interference, and accommodate for the neuromuscular demands of the training. As shown in this example, the consecutive type 1 targets on Tuesday and Wednesday would be likely to limit neuromuscular stress (Tuesday) and facilitate recovery (Wednesday), allowing the hardest work (type 5) to be performed Thursday after the recovery day.

During the NHL season, as with many of the other high-performance team sports, the microcycle

is vastly different (table 26.3). Here, it is important to note that due to the frequency of games (nearly every other day) combined with the travel schedule, it is very challenging to achieve much in the way of quality off-ice training.

For players who are below their time on ice (TOI) thresholds during a home game (forwards 14 to 17 min, defensemen 18 to 22 min), we have them complete additional HIIT work on top of any strength training that may be prescribed on that specific day.

Table 26.3 Example of a Weekly Training Microcycle During the Regular Season

	Sat	Sun	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Focus	Home game	Technical and tactical training on-ice Low-neuro and -metabolic demand Individualized corrective exercise/rehab off-ice	Prepractice strength	Home game	Prepractice individualized corrective exercise/rehab off-ice	Away game	Practice day on the road	Away game On-ice SIT (type 5) in a.m. of game day for players not in lineup that night	Day off
HIIT	SIT (type 5) for players with insufficient TOI			SIT (type 5) for players with insufficient TOI			SIT (Type 5) for players with insufficient TOI in previous game and expected to play insufficient minutes		
Strength	Postgame strength A1) Front squat or trap bar deadlift, work up to 2×3-5 at 70%-80% 1RM A2) Horizontal row, 3×8-12 B1) Single-leg squat, 2×5-7 B2) Single-leg deadlift, 2×5-7 B3) Alternating single-leg push-ups, 2×max		A1) Hang power clean and front squat, 4×1+2 at 1.1 m/s (50%-65% 1RM) or trap bar deadlift jumps, 4×3 at 1.0 m/s A1) Dynamic box squat, 6×3 at 1.1 m/s (55%-65% 1RM+bands) B1) Split stance med ball ballistic throws on wall, 3×5 B2) Glute-ham raise, 3×5	Postgame strength A1) Parallel grip chins, 2-3×4-6 A2) Incline DB press, 2-3×4-6 A3) Rear delt lateral raise, 2-3×8-12 A4) Side bridge, 2-3×5×5 s+30 s isometric					

CASE STUDY

When I initially began training professional ice hockey players in 1998, I had very little experience with the sport. I had never played ice hockey and was only a casual fan. I ran a business training American football players in their off-season, and my training style was heavily influenced by my own personal background competing in American football, powerlifting, and track and field. I slowly began attracting more and more ice hockey players and more or less trained them in the same fashion as I did my football players. I restricted my work solely to the gym assuming (incorrectly) that their coaches would be doing specific conditioning on the ice in addition to their technical and tactical drills. I was always pleasantly surprised at the fact that the athletes I trained would perform at the top of their groups in their team fitness testing despite not having done any of the traditional conditioning that their teammates claimed to have been doing.

I trained my private hockey clients at a track-and-field center that did not have stationary bikes, treadmills, or popular strength-training machines. What we did have was lots of open space, free weights, and medicine balls. We did our strength training with free weights (Olympic lifting and powerlifting), alongside ballistic work with medicine balls and considerable speed and power work on the track.

In 2002, I accepted a position as head strength-and-conditioning coach for the NHL's Toronto Maple Leafs. In addition to being my first full-time strength-and-conditioning position with a high-profile professional team, the position was arguably the most prestigious position in my industry in my country. While I was honored to get the job, I was equally very nervous, as the responsibility came with pressure and scrutiny. In addition to being the youngest person performing this job in the NHL at the time, I was also the least experienced in terms of working with hockey players compared to others in the industry.

The culture of hockey at the time was for teams to perform conditioning programs on the stationary bike after practice to improve their fitness. For example, long slow-duration aerobic rides after games to "clear lactic acid" were the staple prescription (see chapter 3 for discussion on this topic). Most of the weight training that was done was in the form of circuit training, primarily with higher volume and lower intensity protocols. Since everyone was doing this type of training, and I was the least experienced guy in the business, I assumed that I was the one who was ignorant, and that I'd better hedge my bets and get on board with the program.

I spoke with several other strength coaches around the NHL to get the stationary bike programs that they were using; however, after receiving these, the practice still never sat right with me. I could understand the need for "conditioning" in addition to our regular strength training, but could not understand why everyone was performing long- or even medium-length intervals. I heard arguments such as "if shift lengths were 45 s then we should do 45 s intervals," and my favorite was, "if shifts were 45 s long then we should be doing 90 s intervals so that the 45 s shifts feel easy by comparison."

I decided to take it upon myself to study the science of ice hockey. I wanted to really understand the demands of the sport with respect to the underpinning biomechanics and physiology. I began to measure heart rate data during every practice and every game, as well as the players' heart rate variability (HRV; chapter 9) following each game, and the morning following each game, to assess their recovery and readiness to train. We performed comprehensive blood analysis and urinalysis before, during, and after the season to examine measurable effects on the players' biochemistry. While the data collection was very rudimentary, it was still informative. And most importantly, I watched A LOT of hockey.

When I began my investigation in 2002, no available technology provided a reliable way to track the movements and velocities of the players on the ice. This was long before the days of GPS tracking. Our technology was limited to stopwatches and the careful eyes of some volunteer testers, but this was still an extremely useful practice. It did not take long to determine that although a shift may be 45 to 60 s long in total duration, this did not

(continued)

CASE STUDY (CONTINUED)

consist of 45 to 60 s of consistent work, but rather several very short bouts of explosive movements, including accelerations, decelerations, collisions, and shots, interspersed with longer periods of relative inactivity, such as gliding and rest during stoppages in play. This analysis allowed me the confidence to do something drastic and change my practice, going against the grain.

The following season, after some discussion, our coaching staff allowed me to replace our traditional cycle ergometer fitness tests with on-ice tests of speed and endurance, the things that mattered. We chose to test on-ice acceleration and speed with timing gates set up at 0, 20, 35, and 55 m intervals, and the modified Reed repeat sprint skating test (8) to test both speed and recovery ability (and consequently the anaerobic power and aerobic fitness). In addition to timing the athletes, we also collected blood samples from each athlete immediately following the test at specific intervals to determine blood lactate levels (described earlier). Based on the data collected, we were able to create a profile on each player and conduct a needs analysis based on whether they would benefit most from an increased concentration on acceleration, speed, or speed endurance training. The coaching staff also allowed me to replace our traditional steady state aerobic work and long-interval training with more sport-specific sprint interval work. Perhaps an important lesson for future practitioners entering into a new program would be as follows—just because things have always been done a certain way doesn't mean they are right. Make relevant observations, test assumptions, and have the courage to trust not only your data but also your gut and make changes that you truly believe will help your players.